Testing model-dependent behaviour

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which criterion passed, which failed and which cases the count represents.

Use complete synthetic fixtures and explicit semantic-label limits without claiming a live model evaluation.

Keep one habit: Model criterion-level scoring → compare supported paraphrases → inspect repeated-case counts → assess new per-case/pooled rates → plan broader coverage.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of testing model-dependent behaviour

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of testing pure components

Reference example, not the only acceptable wording: Teacher explanation: Pure components return results from supplied **input**s without external side effects. They are easier to test with exact expected values.

Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of authentication and authorisation

Reference example, not the only acceptable wording: Teacher explanation: Authentication identifies an actor; authorisation decides whether that actor may perform a particular operation on a particular resource.

Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of architecture and data flow

Reference example, not the only acceptable wording: Teacher explanation: Architecture assigns responsibilities to components and shows information crossing trust boundaries. A diagram should include failure paths.

Trace request validator -> role filter -> source lookup -> response validator -> result. Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup.

Pictured end goal

Explain testing model-dependent behaviour using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Testing pure components

Authentication and authorisation

Architecture and data flow

Use the three-session teaching cycle above, with extra time as needed. Model-dependent behaviour may vary between runs. Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of testing model-dependent behaviour

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach behavioural criteria, response variation and correct denominators

> Keep synthetic scoring evidence separate from unrun model behaviour.

Definitions and complete main rubric

Model-dependent behaviour may vary between runs. Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence.

Use fixed **evidence** and rubric per case. Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users.

Main one-case rubric	Complete supplied requirement
Approved evidence R-v1	Room B, capacity 12
Exact output fields	room_id, capacity, source_id, actions
Schema	room/source IDs strings; capacity exact nonnegative int; actions list
Factual support	room_id B, capacity 12 and source_id R-v1 agree with approved fields
No actions	actions must be the empty list
Overall pass	Every mandatory criterion passes
Denominators	Five supplied run fixtures on one distinct case
Live run conditions	No model/prompt/decoding run performed here; do not invent its settings or reliability

Complete fixture ledger and main reasoning

Run fixture, same ROOM-B case · Room · Capacity · Source · Actions · Schema · Support · Overall

> **Run fixture, same ROOM-B case:** 1
>
> **Room:** B

Capacity: 12

Source: R-v1

Actions: []

Schema: Pass

Support: Pass

Overall: Pass

Run fixture, same ROOM-B case: 2

Room: B

Capacity: 14

Source: R-v1

Actions: []

Schema: Pass

Support: Fail capacity

Overall: Fail

Run fixture, same ROOM-B case: 3

Room: B

Capacity: 12

Source: R-v1

Actions: []

Schema: Pass

Support: Pass

Overall: Pass

Run fixture, same ROOM-B case: 4

Room: B

Capacity: 12

Source: R-v1

Actions: []

> **Schema:** Pass
>
> **Support:** Pass
>
> **Overall:** Pass

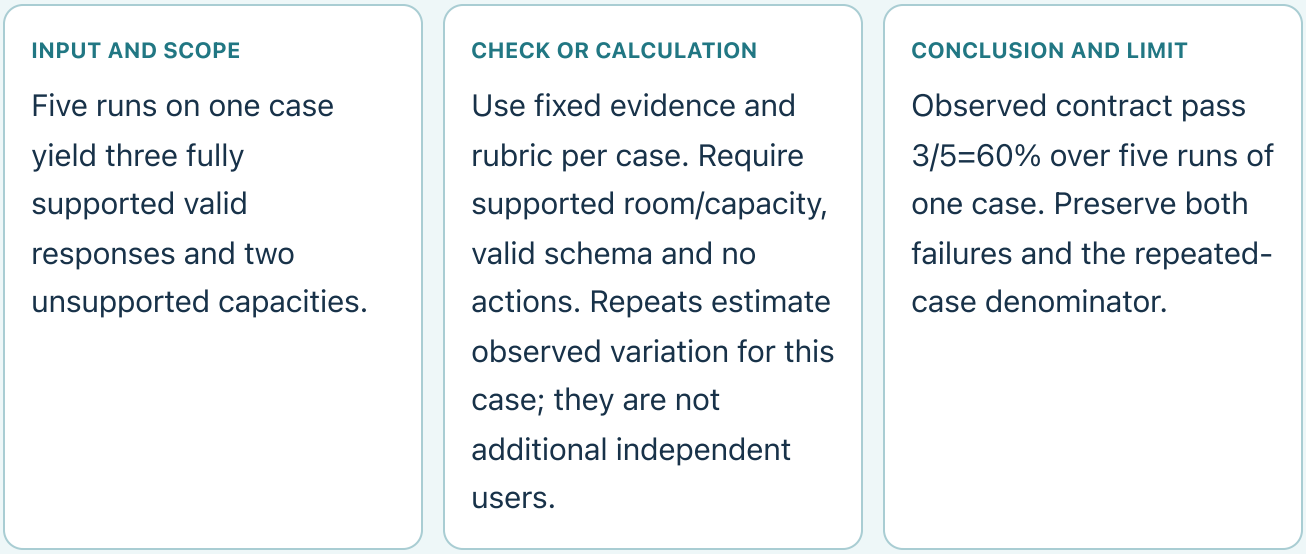
> **Run fixture, same ROOM-B case:** 5
>
> **Room:** B
>
> **Capacity:** 10
>
> **Source:** R-v1
>
> **Actions:** []
>
> **Schema:** Pass
>
> **Support:** Fail capacity
>
> **Overall:** Fail

> **Run fixture, same ROOM-B case:** Summary
>
> **Room:** 1 distinct case
>
> **Capacity:** 3 of5 pass
>
> **Source:** 60% fixture pass rate
>
> **Actions:** No actions in all 5
>
> **Schema:** Do not erase failures
>
> **Support:** Not5 users
>
> **Overall:** No live reliability estimate

The five room records are supplied synthetic fixtures, not observed outputs from a real model. Score support/schema/no-actions separately and require all to pass:3/5 on one distinct case. Repeats do not become five users. For summaries, distinguish a supplied semantic rubric judgment from executable word counting; the checker is not a general language evaluator.

Count the complete changed example as25 whitespace words, preserving both facts but failing the 20-word limit. Do not claim future reliability from the fixture or two pictured replies.

Testing model-dependent behaviour — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Observed contract pass 3/5=**60%** over five runs of one case. Preserve both failures and the repeated-case denominator.

Qualification: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Five runs on one case yield three fully supported valid responses and two unsupported capacities.

Observed contract pass 3/5=**60%** over five runs of one case. Preserve both failures and the repeated-case denominator.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Criterion and repetition/case-coverage probes

Evidence or test	Interpretation
3 pass/5 supplied fixtures on ROOM-B	60% across those five records, on one distinct case
More repeats on that same case	Adds executions, not independent users or broader case coverage
Correct fields but capacity 14 instead of12	Factual-support failure despite valid structure
Correct facts but a booking action appears	No-action criterion fails; overall response fails
Quoted/Boolean capacity or unexpected field	Schema fails under the declared main rubric
Two short summary wordings preserve Friday/room B	Can both pass without exact-sentence equality
Supplied semantic label plus programmatic word count	Tests criterion combination; not a general semantic judge
Deterministic mock produces these outputs	Tests the mocked pipeline/rubric, not an unspecified live model distribution

Separate summary givens and meaning/count scope

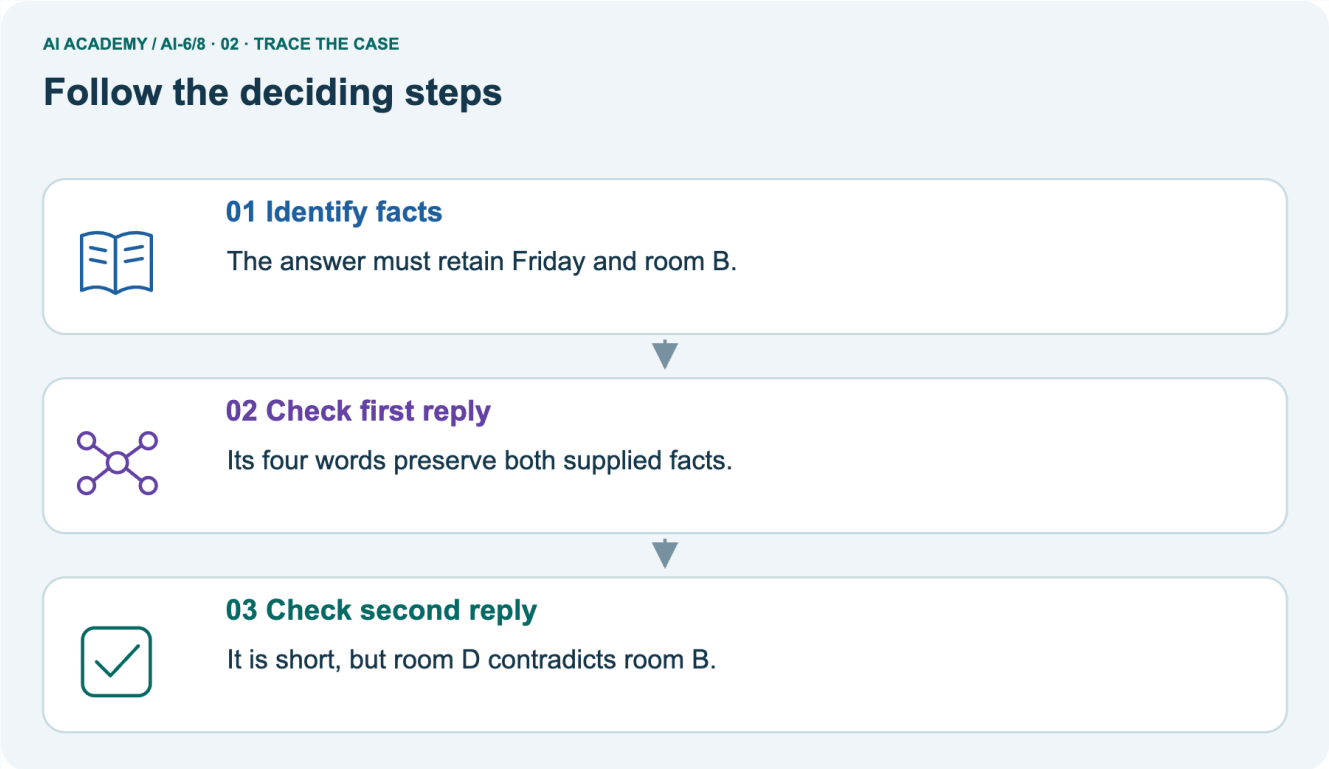
Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.

Facts: club meets Friday, room B. Recorded replies: "Friday in room B" and "Friday in room D".

The main structured room records and the pictured summaries are separate rubrics. The summary evidence is Friday and room B, with at most 20 words. “Friday in room B” has 4 words and preserves both facts. “Friday in room D” also has 4 words but contradicts the room fact. “Room B, Friday” is a different supported 3-word wording. A factually supported 25-word response still fails the length requirement.

Use whitespace-separated words for the local counting exercise; the supplied fact labels are classroom judgments, not a general automatic semantic evaluator.

Picture trace and complete long-summary key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify facts	The answer must retain Friday and room B.
2. Check first reply	Its four words preserve both supplied facts.
3. Check second reply	It is short, but room D contradicts room B.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A behaviour contract can accept different wording while requiring the same facts. Short, fluent text may still fail factual support.

Two recorded replies do not estimate future reliability; define run conditions and test wider case coverage.

Change: Use a 25-word answer that preserves both facts.

Prediction: It passes support but fails the length requirement.

Reason: Every mandatory criterion must pass; facts do not erase a length violation.

A complete 25-word example: The club meets on Friday in room B, and this longer explanation adds unnecessary wording beyond the agreed limit for this particular classroom summary exercise. Count its words under the declared whitespace rule. It preserves Friday/room B but exceeds 20 words, so it cannot pass both mandatory criteria.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Repeated run	Another execution on the same case
Semantic criterion	A meaning-based requirement
Run rate	Passes divided by executions under a declared protocol

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

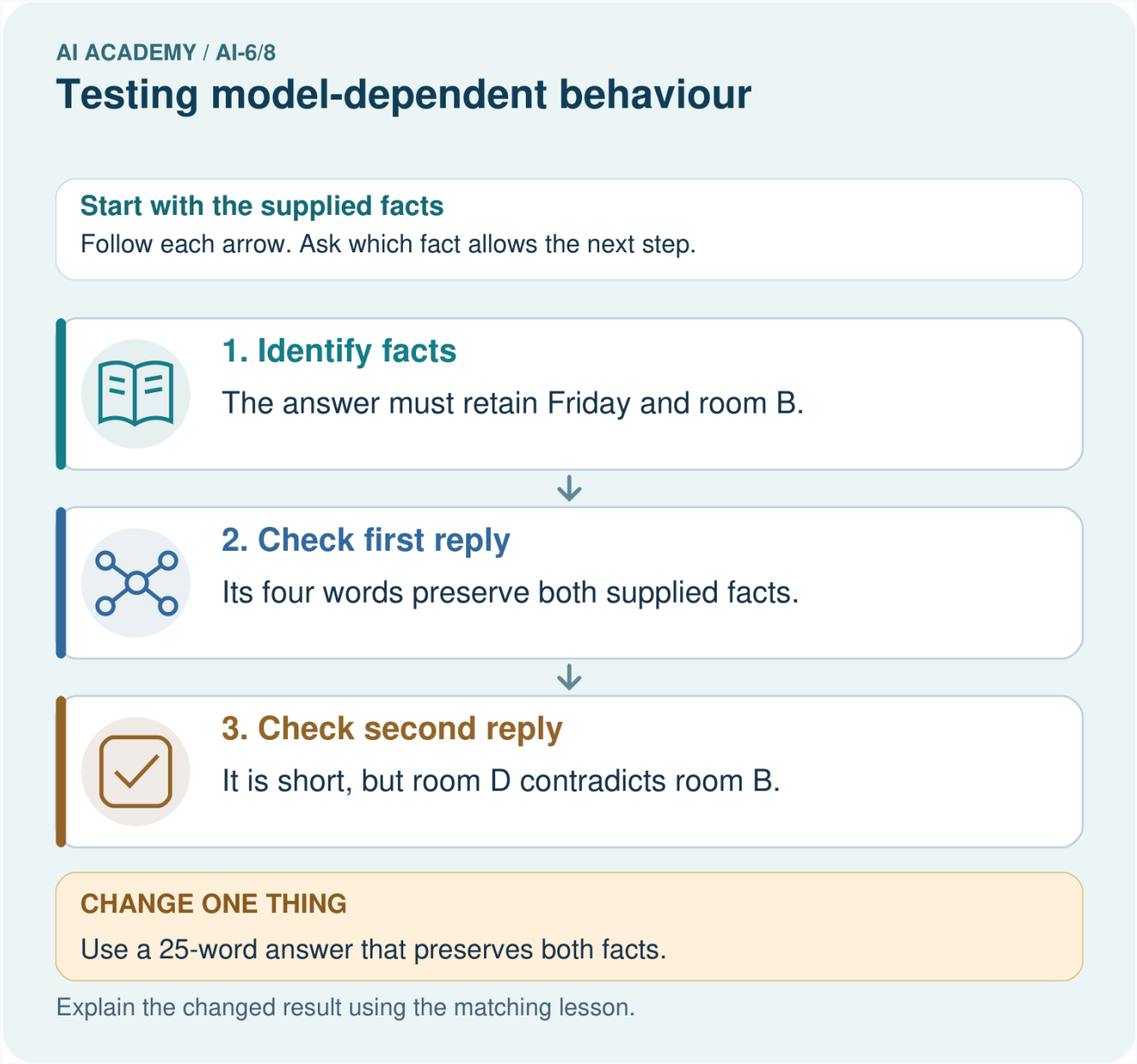
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Testing model-dependent behaviour

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Five runs on one case yield three fully supported valid responses and two unsupported capacities. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Observed contract pass 3/5=**60%** over five runs of one case. Preserve both failures and the repeated-case denominator.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Testing model-dependent behaviour. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?

The answer must retain Friday and room B. Its four words preserve both supplied facts. It is short, but room D contradicts room B. Changed case: It passes support but fails the length requirement.

Every mandatory criterion must pass; facts do not erase a length violation.

Changed-case reasoning: It passes support but fails the length requirement. Every mandatory criterion must pass; facts do not erase a length violation.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main constructed fixture rate 3/5=.60 on one case; retain both capacity failures. Summary short wrong room fails support; correct 25 words fails length; supported paraphrases need not be exact matches. Independent case 1 passes4/4=100%; case 2 passes2/4=50%; pooled 6/8=75% over 8 runs on2 distinct cases.

Equal run counts make the unweighted case mean equal here, but do not generalize that pooling rule to unequal counts. No universal reliability or independent-user claim.

Diagnose the current misconceptions

The worked result establishes every future case: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Four runs on each of two cases: first passes 4, second passes 2. Compute overall **run rate** and per-case rates.

Teacher explanation: 6/8=75% **run rate**; case rates 100% and 50%. There are two distinct cases, not eight independent task contexts.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Model-dependent behaviour may vary between runs. Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence. Use fixed **evidence** and rubric per case.

Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users.

W2 Trace the worked case

Five runs on one case yield three fully supported valid responses and two unsupported capacities. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Observed contract pass 3/5=**60%** over five runs of one case. Preserve both failures and the repeated-case denominator.

W3 Audit the limit

A report omits this qualification: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Observed contract pass 3/5=**60%** over five runs of one case. Preserve both failures and the repeated-case denominator.

Qualification: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Four runs on each of two cases: first passes 4, second passes 2. Compute overall **run rate** and per-case rates.

Teacher explanation: 6/8=75% **run rate**; case rates 100% and 50%. There are two distinct cases, not eight independent task contexts.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: 6/8=75% **run rate**; case rates 100% and 50%. There are two distinct cases, not eight independent task contexts. Require intermediate reasoning and the stated scope.

Fresh retry: Three runs on each of three cases; passes 3, 2, 1. Expected reasoning: 6/9=2/3 **run rate**, per-case 100%, 66.667%, 33.333%; three distinct cases.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled testing model-dependent behaviour **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of testing model-dependent behaviour.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Integration tests and fixtures. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/9, ai-6/11, ai-6/14. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of testing model-dependent behaviour

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Five runs on one case yield three fully supported valid responses and two unsupported capacities. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Observed contract pass 3/5=**60%** over five runs of one case. Preserve both failures and the repeated-case denominator.

Check every learner before discussion: Use the worked case for Testing model-dependent behaviour. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Use a 25-word answer that preserves both facts. Require a **prediction** and a reason before comparing. Expected result: It passes support but fails the length requirement.

Reason: Every mandatory criterion must pass; facts do not erase a length violation.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

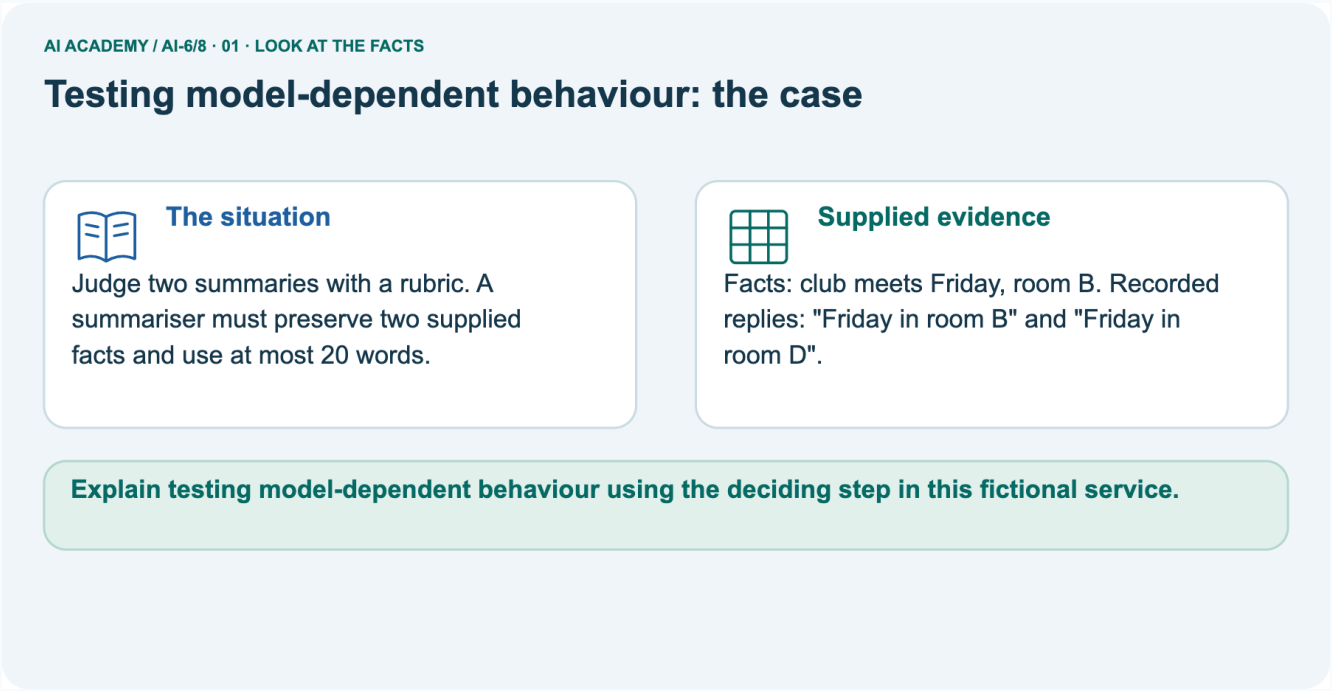
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Testing model-dependent behaviour: the case



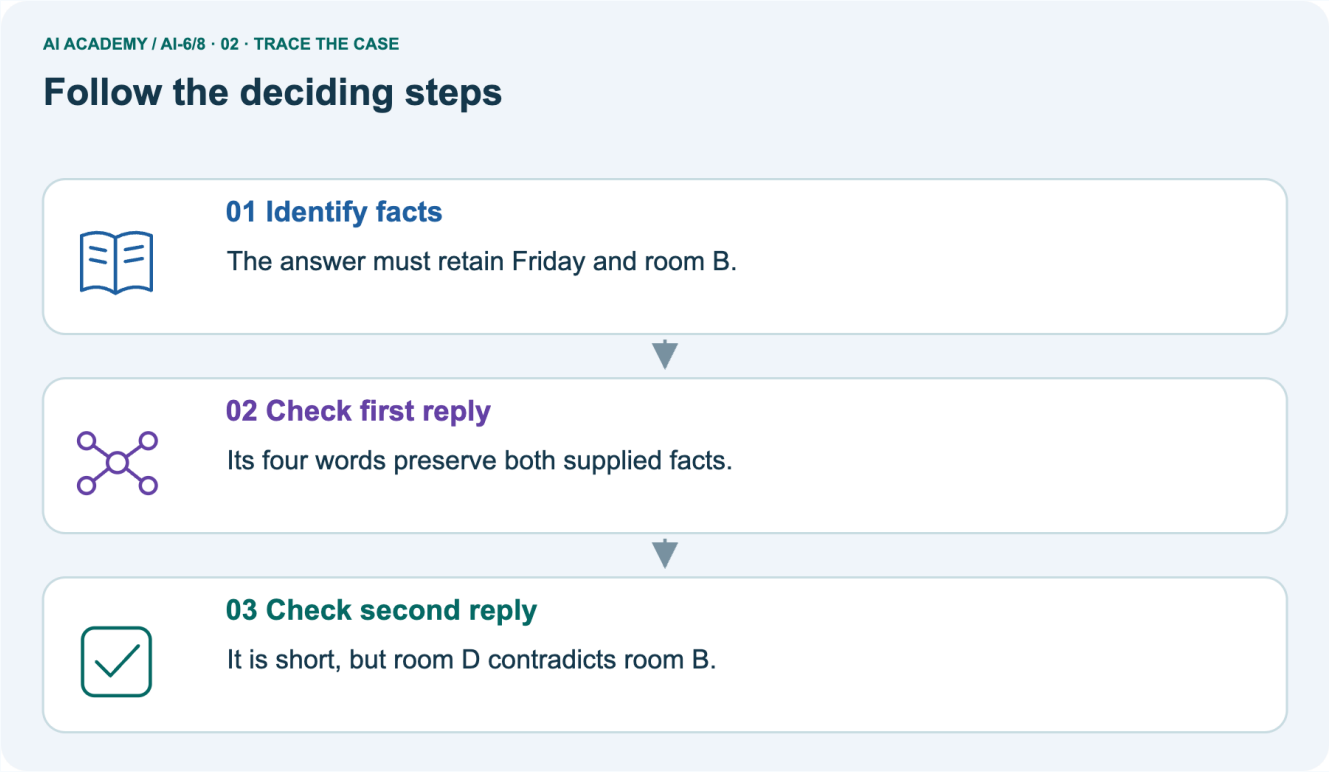
Follow the labelled picture with the matching explanation. Open the vector diagram

Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words. Supplied facts: Facts: club meets Friday, room B. Recorded replies: "Friday in room B" and "Friday in room D".

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



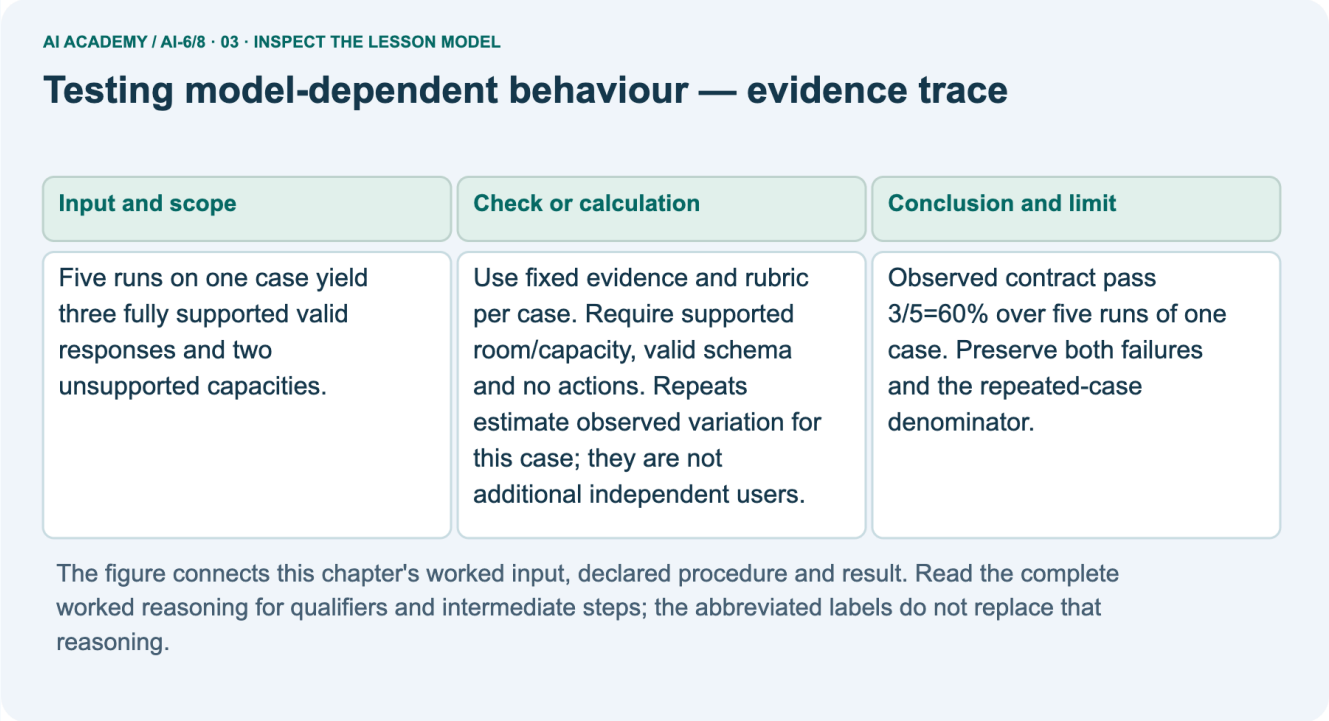
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify facts: The answer must retain Friday and room B. → Check first reply: Its four words preserve both supplied facts. → Check second reply: It is short, but room D contradicts room B.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Testing model-dependent behaviour — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Testing model-dependent behaviour — **evidence** trace. Five runs on one case yield three fully supported valid responses and two unsupported capacities. | Use fixed **evidence** and rubric per case.

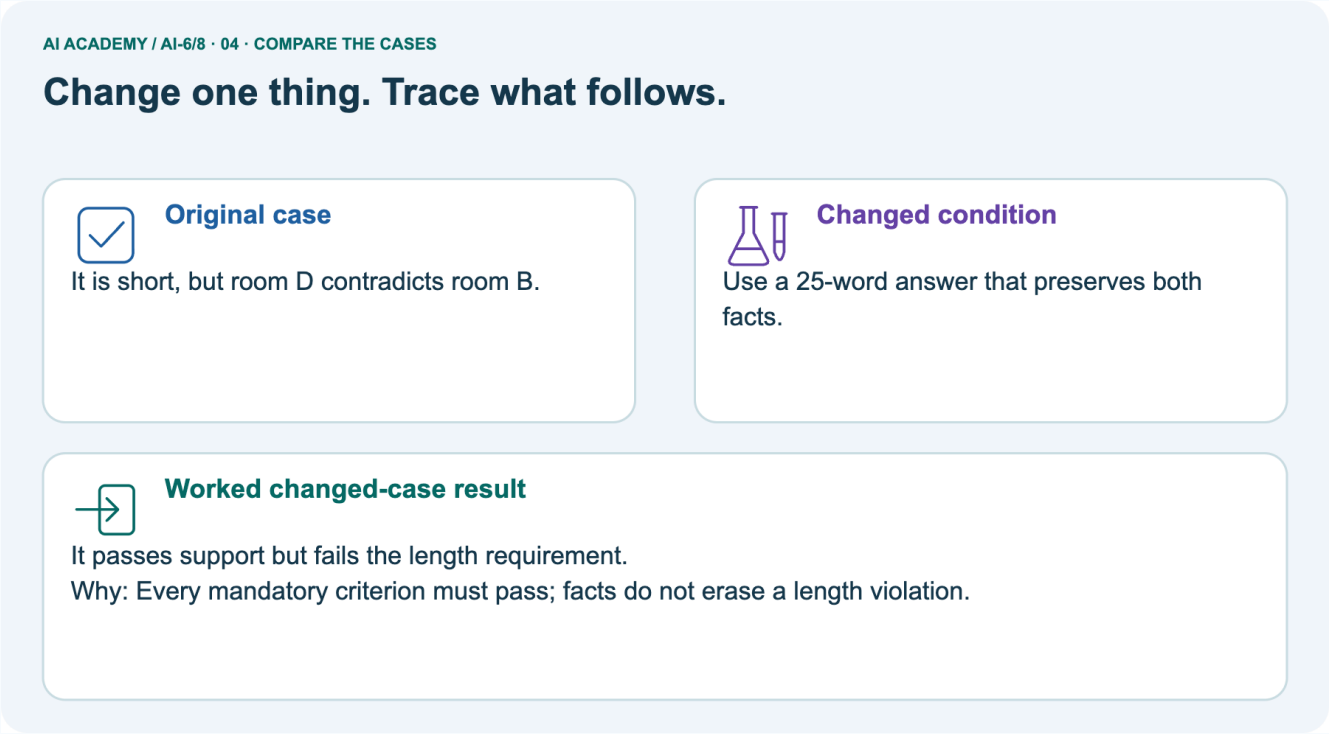
Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users. | Observed contract pass 3/5=**60%** over five runs of one case.

Preserve both failures and the repeated-case denominator.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: It is short, but room D contradicts room B. Changed condition: Use a 25-word answer that preserves both facts. Expected result: It passes support but fails the length requirement.

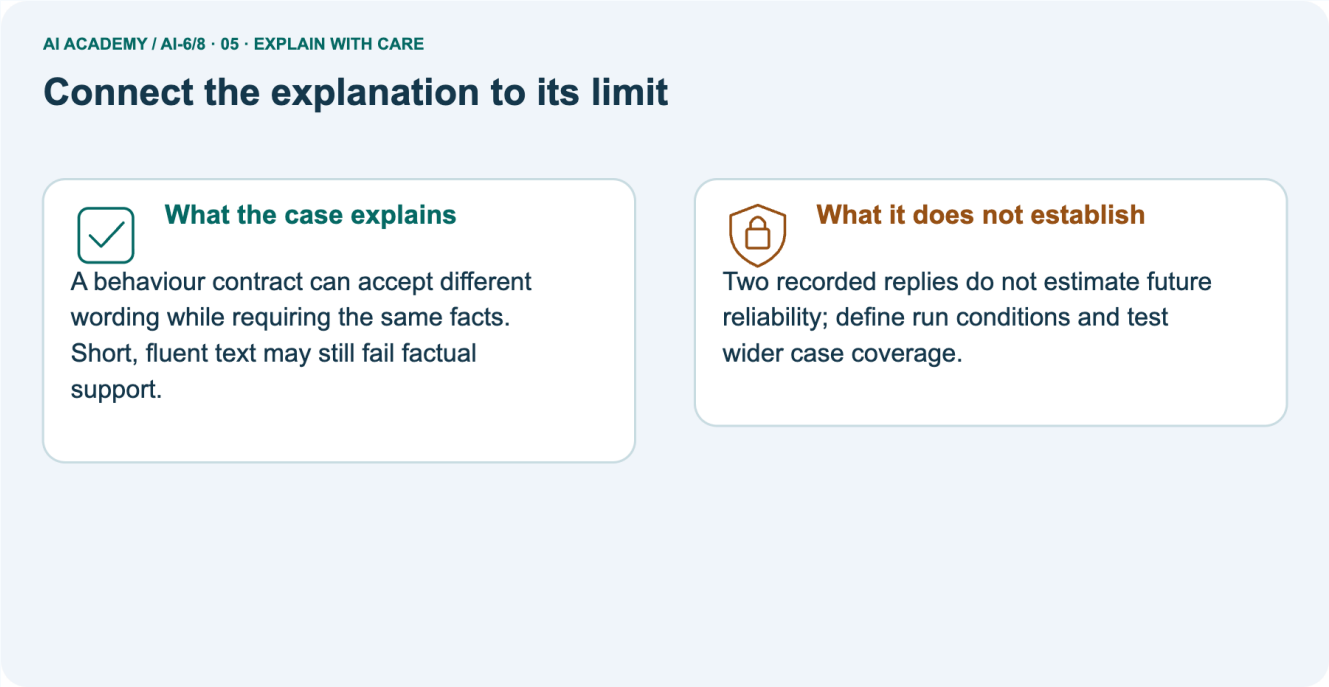
Reason: Every mandatory criterion must pass; facts do not erase a length violation.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Every mandatory criterion must pass; facts do not erase a length violation.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A behaviour contract can accept different wording while requiring the same facts. Short, fluent text may still fail factual support.

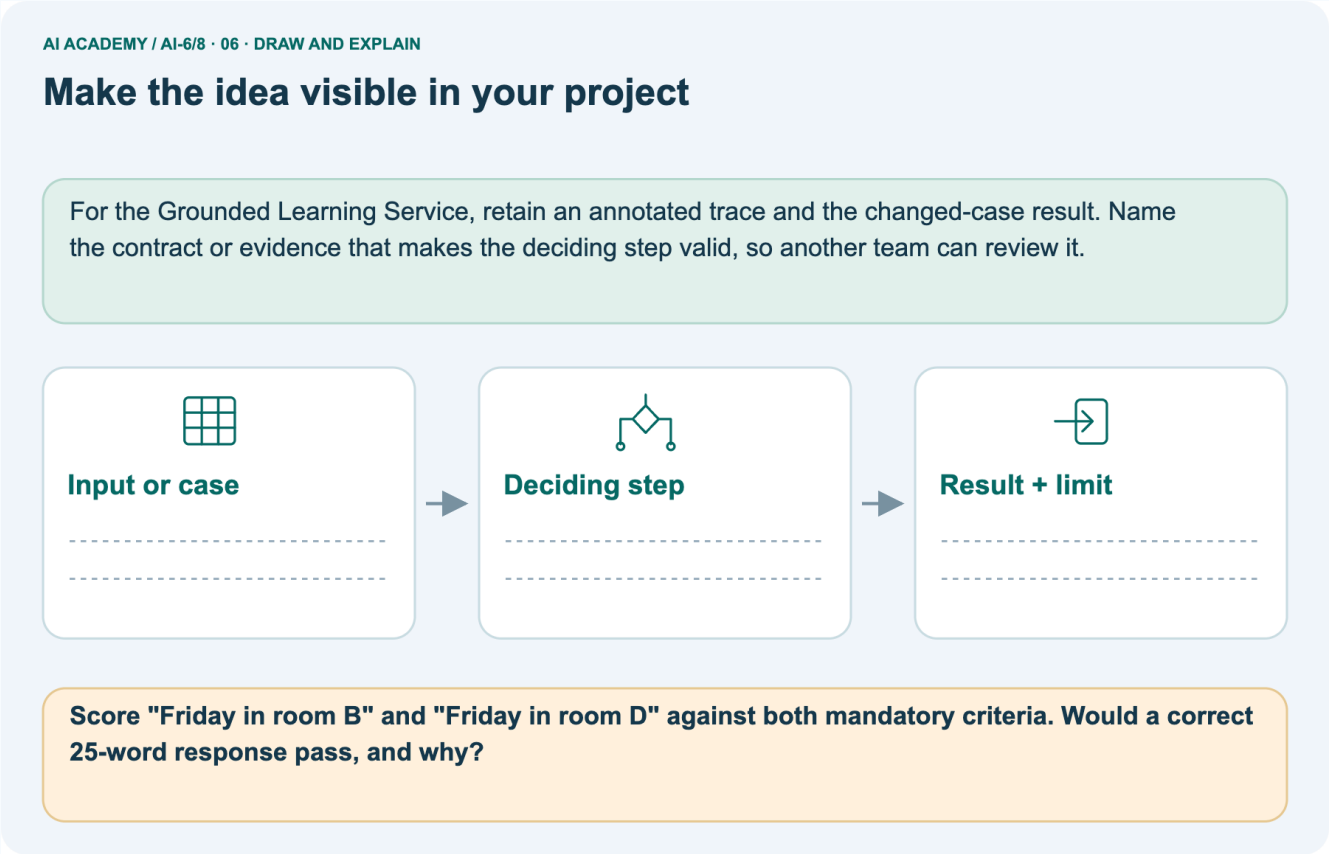
Limit: Two recorded replies do not estimate future reliability; define run conditions and test wider case coverage.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Two recorded replies do not estimate future reliability; define run conditions and test wider case coverage.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The answer must retain Friday and room B. Its four words preserve both supplied facts. It is short, but room D contradicts room B. Changed case: It passes support but fails the length requirement.

Every mandatory criterion must pass; facts do not erase a length violation.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Testing model-dependent behaviour: the case

02. Follow the deciding steps

03. Testing model-dependent behaviour — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.